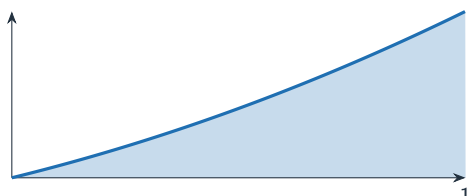
Условие

Дадена е X с плътност $f_X(x) = c(x^2 + 2x)\mathbb{1}_{[0,1]}(x)$. Намерете (а) c; (б) $\mathbb{E}[X]$ и $\mathrm{Var}(X)$; (в) $\mathbb{P}(X \leq \mathbb{E}[X])$; (г) $\mathbb{E}[X^2 + 3X]$.

(а) Константата c

Плътността трябва да има лице 1:

$$1 = \int_0^1 c(x^2 + 2x)\,dx = c\Big[\frac{x^3}{3} + x^2\Big]_0^1 = c\Big(\frac{1}{3} + 1\Big) = \frac{4}{3}c \quad \Longrightarrow \quad c = \frac{3}{4}.$$



$f_X(x) = \frac{3}{4}(x^2 + 2x)$, лице 1

Проверяваме и $f_X \geq 0$: на $[0, 1]$ и x^2, и $2x$ са неотрицателни. ✓

(б) Очакване и дисперсия

$$\mathbb{E}X = \int_0^1 x \cdot \frac{3}{4}(x^2 + 2x)\,dx = \frac{3}{4}\int_0^1 (x^3 + 2x^2)\,dx = \frac{3}{4}\Big(\frac{1}{4} + \frac{2}{3}\Big) = \frac{3}{4} \cdot \frac{11}{12} = \frac{11}{16}.$$

$$\mathbb{E}[X^2] = \frac{3}{4}\int_0^1 (x^4 + 2x^3)\,dx = \frac{3}{4}\Big(\frac{1}{5} + \frac{1}{2}\Big) = \frac{3}{4} \cdot \frac{7}{10} = \frac{21}{40}.$$

$$\mathrm{Var}X = \frac{21}{40} - \Big(\frac{11}{16}\Big)^2 = \frac{21}{40} - \frac{121}{256} = \frac{672 - 605}{1280} = \frac{67}{1280} \approx 0{,}052.$$

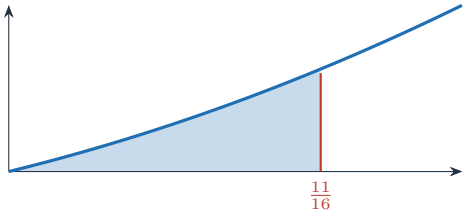
Плътността расте, затова масата е изместена надясно: $\mathbb{E}X \approx 0{,}69 > 0{,}5$.

(в) $\mathbb{P}(X \leq \mathbb{E}X)$

Функцията на разпределение за $x \in [0,1]$:

$$F_X(x) = \frac{3}{4}\Big(\frac{x^3}{3} + x^2\Big).$$

$$\mathbb{P}\Big(X \leq \frac{11}{16}\Big) = \frac{3}{4}\Big(\frac{(11/16)^3}{3} + \Big(\frac{11}{16}\Big)^2\Big) \approx \frac{3}{4}\big(0{,}1083 + 0{,}4727\big) \approx 0{,}436.$$



защриховано: $\approx 0{,}436$

При несиметрично разпределение $\mathbb{P}(X \leq \mathbb{E}X)$ не е задължително $\frac{1}{2}$.

(г) $\mathbb{E}[X^2 + 3X]$

По линейност на очакването:

$$\mathbb{E}[X^2 + 3X] = \mathbb{E}[X^2] + 3\,\mathbb{E}X = \frac{21}{40} + 3 \cdot \frac{11}{16} = \frac{42}{80} + \frac{165}{80} = \frac{207}{80} \approx 2{,}59.$$

Отговор. (а) $c = \frac{3}{4}$; (б) $\mathbb{E}X = \frac{11}{16}$, $\mathrm{Var}X = \frac{67}{1280}$; (в) $\approx 0{,}436$; (г) $\frac{207}{80}$.